

DHINESH KUMAR

+91 8056711549 | dhineshkumarprakasam@gmail.com | [Github](#) | [LinkedIn](#) | [LeetCode](#)

SUMMARY

Software engineer with experience in backend development, Python automation, and system optimization. Focused on building maintainable, scalable applications through efficient code design and data-driven problem solving. Proficient in integrating AI-based workflows and managing backend infrastructure to support performant software solutions.

EDUCATION

Master of Computer Applications - **10 CGPA (University Rank : 1)**

VIT, Vellore - July 2026 – July 2028

Bachelor of Science in Computer Science (Data Science) - **9.47 CGPA (University Rank : 4)**

VIT, Vellore - July 2023 – July 2026

SKILLS

- **Programming Languages** : Python, C++, Java
- **Backend Development** : FastAPI, Flask, Spring Boot, Node.js, Express.js, REST APIs
- **Frontend & Mobile** : React.js, Flutter, Bootstrap, jQuery, HTML5, CSS3
- **Databases & Caching** : MySQL, MongoDB, Redis
- **Data Science** : Machine Learning Predictive Modeling (Sklearn), Data Wrangling (Pandas), Visualization(Matplotlib, Seaborn)
- **GenAI & LLMs** : Retrieval-Augmented Generation(RAG), ChromaDB
- **Developer Tools** : Linux (CLI), Docker, Git, Github Actions, Postman

PROJECTS

JQL (Json Query Language) | [Github](#)

System Software (Python, JavaScript, Bootstrap, HTML, CSS)

- **Architected a lightweight, standalone desktop application (CLI & GUI)** that lets users query, search, and modify raw JSON files instantly without installing heavy databases like MySQL or MongoDB.
- **Built an interactive CLI shell using python `prompt_toolkit`** with real-time query validation and auto-completion, providing instant error feedback mid-typing to block broken commands before execution.
- **Integrated an LRU in-memory cache**, slashing average file-retrieval latency by **~97%** (from **13.02 ms** without cache to **0.35 ms** with cache).

HelpMe@OS | [Github](#) | [Live Demo](#)

Web Application (Python, Flask, ChromaDB Cloud Client, Google Gemini AI)

- **Developed an interactive OS simulation platform** featuring **step-by-step solvers** and **dynamic visualizations** for 18 classical algorithms across CPU scheduling, memory management, disk scheduling, and page replacement.
- **Engineered a RAG-based AI assistant** using **Gemini AI** and **ChromaDB** (4,732 chunks) to provide context-aware technical support from OS documentation, optimized with **rate-limiting** for scalable performance

D - Secure Vault | [Github](#)

Mobile Application (Flutter, Dart, Kotlin)

- **Developed a secure Android app** encrypting PDFs into a proprietary `.fsbd` format, forcing a **strictly view-only environment** with zero ability to save or store the decrypted file.
- **Integrated Kotlin native** protections to block screenshots/recordings and instantly wipe temporary decrypted files from memory upon exit.

EXPERIENCE

Lysa Solutions (Remote) – Python Developer Intern

June 2025 – July 2025

- **Developed an end-to-end automated pipeline** to scrape, download, and parse **500+** previous year question papers, utilizing **intelligent diagram detection** to extract and structure data into a comprehensive dataset.
- **Streamlined** the development of a **GenAI question paper generator** by transforming raw PDFs into a clean, structured **Excel dataset**, significantly reducing manual data preparation time.

HONORS & INVOLVEMENT

- Secured **3rd place** in **Python Code-a-thon** organized by Capsule Labs.
- Awarded Merit Scholarship from VIT for **consistent academic excellence over 3 consecutive years**.
- Solved over **390+** programming problems in **Leetcode**